

2.5 Normalization

Step 1: Unnormalized Form (UNF)

In the beginning, all data may exist in one large table containing repeating groups and redundant information.

UNF Table

student_id	student_name	department_name	course_id	course_name	instructor_name	enrollment_date	grade
101	Abel Tadesse	Computer Science	C101	Database Systems	Dr. Bekele	2026-01-10	A
101	Abel Tadesse	Computer Science	C102	Networking	Dr. Hana	2026-01-10	B
102	Meron Ali	Software Engineering	C101	Database Systems	Dr. Bekele	2026-01-11	A

Problems in UNF

- Repeating course records for one student
- Data redundancy
- Difficult update operations
- Insertion and deletion anomalies
- Poor data consistency

First Normal Form (1NF)

A table is in 1NF when:

- All attributes contain atomic (single) values
- No repeating groups exist
- Each row is unique

1NF Relation

enrollme nt_id	studen t_id	student _name	departmen t_name	cours e_id	course_n ame	instruc tor_nam e	enrollmen t_date	grade
E001	101	Abel Tadesse	Computer Science	C101	Database Systems	Dr. Bekele	2026-01- 10	A
E002	101	Abel Tadesse	Computer Science	C102	Networki ng	Dr. Hana	2026-01- 10	B
E003	102	Meron Ali	Software Engineeri ng	C101	Database Systems	Dr. Bekele	2026-01- 11	A

Improvements Achieved

- Repeating groups removed
- Atomic values ensured
- Easier searching and sorting

Remaining Problems

- Partial dependency still exists
- Data redundancy remains
- Update anomalies still possible

Second Normal Form (2NF)

A table is in 2NF when:

- It is already in 1NF
- All non-key attributes fully depend on the entire primary key
- No partial dependency exists

To achieve 2NF, the large table is decomposed into separate relations.

STUDENT Table

student_id (PK)	first_name	last_name	email	phone	date_of_birth	department_id
101	Abel	Tadesse	abel@gmail.com	0911223344	2003-04-12	D01
102	Merlon	Ali	meron@gmail.com	0911334455	2002-08-21	D02

DEPARTMENT Table

department_id (PK)	department_name
D01	Computer Science
D02	Software Engineering

INSTRUCTOR Table

instructor_id (PK)	first_name	last_name	email	department_id
I01	Bekele	Alemu	bekele@aastu.edu.et	D01
I02	Hana	Tesfaye	hana@aastu.edu.et	D02

COURSE Table

course_id (PK)	course_name	credit_hour	instructor_id
C101	Database Systems	3	I01
C102	Networking	4	I02

ENROLLMENT Table

enrollment_id (PK)	student_id (FK)	course_id (FK)	enrollment_date	grade
E001	101	C101	2026-01-10	A
E002	101	C102	2026-01-10	B

enrollment_id (PK)	student_id (FK)	course_id (FK)	enrollment_date	grade
E003	102	C101	2026-01-11	A

Improvements Achieved

- Partial dependencies removed
- Reduced redundancy
- Better organization of data

Remaining Problem

- Possible transitive dependencies

Third Normal Form (3NF)

A relation is in **3NF** if:

- It is already in 2NF
- No transitive dependency exists
- Non-key attributes depend only on the primary key

3NF Tables

1. DEPARTMENT Table (3NF)

department_id (PK)	department_name
1	Computer Science
2	Software Engineering
3	Information Technology

2. STUDENT Table (3NF)

student_id (PK)	first_name	last_name	email	phone	date_of_birth	department_id (FK)
101	Abel	Tadesse	abel@gmail.com	0911223344	2003-04-12	1

student_id (PK)	first_name	last_name	email	phone	date_of_birth	department_id (FK)
102	Merlon	Ali	meron@gmail.com	0911334455	2002-08-21	2

3. INSTRUCTOR Table (3NF)

instructor_id (PK)	first_name	last_name	email	department_id (FK)
201	Bekele	Alemu	bekele@aastu.edu.et	1
202	Hana	Tesfaye	hana@aastu.edu.et	2

4. COURSE Table (3NF)

course_id (PK)	course_name	credit_hour	instructor_id (FK)
301	Database Systems	3	201
302	Networking	4	202

5. ENROLLMENT Table (3NF)

enrollment_id (PK)	student_id (FK)	course_id (FK)	enrollment_date	grade
401	101	301	2026-01-10	A
402	101	302	2026-01-10	B
403	102	301	2026-01-11	A

Boyce-Codd Normal Form (BCNF)

A relation is in **BCNF** if:

- For every functional dependency:

$X \rightarrow Y \mid X \twoheadrightarrow Y \mid X \twoheadrightarrow Y \mid X \twoheadrightarrow Y$

- X must be a candidate key.

BCNF Tables

1. DEPARTMENT Table (BCNF)

department_id (PK)	department_name
1	Computer Science
2	Software Engineering
3	Information Technology

Functional Dependency:

department_id → department_name
 department_name → department_id

2. STUDENT Table (BCNF)

student_id (PK)	first_name	last_name	email	phone	date_of_birth	department_id (FK)
101	Abel	Tadesse	abel@gmail.com	0911223344	2003-04-12	1
102	Meron	Ali	meron@gmail.com	0911334455	2002-08-21	2

Functional Dependency:

student_id → first_name, last_name, email, phone, date_of_birth, department_id
 first_name, last_name, email, phone, date_of_birth → student_id
 department_id → first_name, last_name, email, phone, date_of_birth, department_id

3. INSTRUCTOR Table (BCNF)

instructor_id (PK)	first_name	last_name	email	department_id (FK)
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instructor_id (PK)	first_name	last_name	email	department_id (FK)
201	Bekele	Alemu	bekele@aastu.edu.et	1
202	Hana	Tesfaye	hana@aastu.edu.et	2

Functional Dependency:

$\text{instructor_id} \rightarrow \text{first_name}, \text{last_name}, \text{email}, \text{department_id}$
 $\text{first_name}, \text{last_name}, \text{email} \rightarrow \text{department_id}$
 $\text{department_id} \rightarrow \text{instructor_id}$

4. COURSE Table (BCNF)

course_id (PK)	course_name	credit_hour	instructor_id (FK)
301	Database Systems	3	201
302	Networking	4	202

Functional Dependency:

$\text{course_id} \rightarrow \text{course_name}, \text{credit_hour}, \text{instructor_id}$
 $\text{course_name}, \text{credit_hour} \rightarrow \text{instructor_id}$
 $\text{instructor_id} \rightarrow \text{course_id}$

5. ENROLLMENT Table (BCNF)

enrollment_id (PK)	student_id (FK)	course_id (FK)	enrollment_date	grade
401	101	301	2026-01-10	A
402	101	302	2026-01-10	B
403	102	301	2026-01-11	A

Functional Dependency:

$\text{enrollment_id} \rightarrow \text{student_id}, \text{course_id}, \text{enrollment_date}, \text{grade}$
 $\text{student_id}, \text{course_id}, \text{enrollment_date} \rightarrow \text{grade}$
 $\text{grade} \rightarrow \text{enrollment_id}$

